

Etienne MARION

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I am studying Mathematics at École Normale Supérieure de Lyon. I completed my M.Sc. in probability and statistics and am now doing research internships before starting my PhD next year. I am mainly interested in probability theory applied to population genetics and formalization of Mathematics using the proof assistant Lean.

Education

- **2025-2026:** Research internships before starting my PhD :
Feb - Jul 2026: intenship at Université Paris-Cité supervised by Félix Foutel-Rodier;
Sep 2025 - Jan 2026: internship at the university of Vienna supervised by Emmanuel Schertzer.
- **2023-2025:** M.Sc. in probability and statistics at École Normale Supérieure de Lyon.
- **2022-2023:** B.Sc. in Mathematics at École Normale Supérieure de Lyon.
- **2020-2022:** Classe préparatoire aux grandes écoles at Lycée Georges Clemenceau in Reims.

Research experience

- **Jul 2026:** I mam invited to INRIA by Rémy Degenne to work on formalizing stochastic integrals in Lean.
- **Feb - Jul 2026:** Internship at Université Paris-Cité, supervised by Félix Foutel-Rodier : *Impact of demographic fluctuations sur structured populations genealogies.*
- **Sep 2025 - Jan 2026:** Internship at the University of Vienna, supervised by Emmanuel Schertzer : *Diffusive neutral fractions inside a genetic population.*
- **Apr - Jul 2025:** Internship at École Normale Supérieure de Lyon, supervised by Jean-Christophe Mourrat : *Study of functions learnable by neural networks using mean-fields approximation..*
- **May - Jul 2024:** Internship at Institut de recherche mathématique de Rennes, supervised by Sébastien Gouëzel : *Formalization of the Ionescu-Tulca theorem and construction of infinite products of porobability measures with Lean.*
- **May - Jun 2023:** Internship at laboratoire Jean Kuntzmann in Grenoble, supervised by Sana Louhichi : *Tight sequences of random variables: definition, examples and application.*

Preprints

- Rémy Degenne, David Ledvinka, Etienne Marion, Peter Pfaffelhuber (2025+)
Formalization of Brownian motion in Lean
<https://arxiv.org/abs/2511.20118>

- Etienne Marion (2025+)
A Formalization of the Ionescu-Tulcea Theorem in Mathlib
<https://arxiv.org/abs/2506.18616>

Talk

- *A Formalization of the Ionescu-Tulcea Theorem in Mathlib*
 Lean Together 2026
https://www.youtube.com/watch?v=_TrTKnWU-to

Participations to conferences

- **Apr 2026:** I was present at the 3rd annual meeting of the GdR Branchement in Avignon.
- **Sep 2025:** I was present at the workshop Ancestral lines in population models with interaction at BOKU University, in Vienne.
- **Jun 2025:** I was present at the summer school Local Limits of Random Graphs at Institut de Mathématique d'Orsay.
- **Mar 2024:** I was present at the conference Lean for the curious mathematician at CIRM.

Other activities

- I am part of the Mathlib reviewer team. I am assessing the mathematical relevance and the quality of the code that contributors wish to add to the library, especially in the probability section.
- **Sep - Dec 2023:** With three fellow students we gave an introductory course to category theory, shaped as a weekly seminar.